ASIDE: An Agentic Self-Improving Data Engine for Domain-Specific Visual Annotation

Efe Aydeşkin

Technical Report / Research Proposal — Version 0.1
September 2026

Abstract

Industrial computer-vision projects are rarely bottlenecked by training YOLO or DETR; they are bottlenecked by producing enough correctly labeled data. We present **ASIDE** (Agentic Self-Improving Data Engine), a research proposal and system architecture that aims to reduce—not eliminate—human annotation cost for domain-specific object detection. ASIDE combines public-dataset discovery, ontology alignment, schema normalization, sequence-based annotation, retrieval-augmented visual memory, teacher-model consensus, active learning, and downstream detector feedback into a closed loop.

A sequence detector alone is not claimed as novelty: Pix2Seq already casts detection as discrete token generation [2]; Florence-2 unifies vision tasks under seq2seq with large-scale automated annotation [7]; Grounding DINO provides open-set detection [4]; and SAM 2 supplies promptable segmentation with temporal memory [5]. The combination we propose to study is sequence annotation with semantic retrieval memory, temporal memory, error memory, model consensus, active learning, and closed-loop detector feedback. Two design principles organize the system: (i) confidence is not correctness—acceptance uses multi-signal reliability; (ii) not everything should be an agent—query expansion may be agentic, while format conversion and hashing must be deterministic. Version 0.1 scopes the first experiment to a small Image→annotation-sequence model; VectorDB and consensus are measurable ablation targets for later versions. This document is a working research proposal; it is not a peer-reviewed paper and reports no trained model or measured human-hour experiment.

Primary research question. Can retrieval-augmented visual memory, interconnected visual sequences, model disagreement, and learning-history signals reduce human labeling load for domain-specific object detection?

Claim boundary for v0.1. Formulas are definitional. Figures that show e.g. "80 h → 8 h" are target scenarios, not experimental results. We do *not* claim invention of sequence detection, replacement of Pix2Seq/Florence-2/Grounding DINO/SAM 2, fully automatic labeling, a measured 90% time reduction, verified novelty, or demonstrated factory-scale success.

Contents

1 Purpose and scope **3**

2 Problem: the labeled-data bottleneck **3**

3 Central hypothesis: from human annotation to a self-improving engine **3**

4 Sequence detection and literature boundary **3**

5 System overview **4**

6 Task Agent and the agentic / deterministic split **4**

7 Dataset discovery, registry, and quality **4**

8 Ontology Agent **4**

9 Annotation schema normalization **4**

10 Vision encoder **5**

11 Annotation tokenizer and sequence format **5**

12 EOS, variable length, and object ordering **5**

13 Sequence model and loss **5**

14 Pure Seq2Seq versus hybrid geometry **6**

15 Linked sequences and temporal memory **6**

16 Production sequence memory **6**

17 VectorDB and Visual RAG **6**

18 Four memory types and retrieval attention **6**

19 Core formula: short and long memory **7**

20 Storage architecture **7**

21 Teacher models, SAM 2, and consensus **7**

22 Humans, active learning, and error memory **7**

23 Hard negatives and the self-improving loop **8**

24 Domain shift and customer data **8**

25 Rich outputs, relations, and the role of YOLO **8**

26 Relation to prior work **8**

27 Research question, originality, and ablation **8**

28 Version plan and the v0.1 experiment **9**

29 Evaluation and success criteria **9**

30 Conclusion **10**

A Symbol glossary **10**

B Agents and services **10**

C Selected reference anchors **11**

1 Purpose and scope

Version 0.1 does three things: (i) state the hypothesis mathematically; (ii) record the target architecture and memory-binding formula; (iii) separate the scientific test order from claims that cannot yet be made. This document is not an implementation guide or production API specification; application frameworks, queues, and UI layers are deliberately out of scope.

Explicit non-claims. We do not claim that we invented object detection as a sequence; that we replace Pix2Seq, Florence-2, Grounding DINO, or SAM 2; that labeling is 100% automatic; that human labeling time has been measured to drop by 90% (that is a goal); that novelty is already verified; or that the method works on large models or in a real factory.

2 Problem: the labeled-data bottleneck

Training YOLO [6], DETR [1], or another CV model is often not the hardest part of an industrial project. The bottleneck is producing correct and sufficient labeled data. In a marble factory seeking cracks, holes, cavities, veins, discoloration, and surface defects, tens or hundreds of thousands of customer images may need class, box, and optionally mask labels.

> *One-line summary:* the bottleneck is labeled-data production, not model architecture.

3 Central hypothesis: from human annotation to a self-improving engine

The goal is not merely an auto-labeling script. It is to turn human labeling into a machine-assisted, self-correcting annotation process. Humans are not removed; cost is reduced. The target regime is one in which humans touch only uncertain and novel examples.

This expands the project beyond a single detector into an agentic self-improving CV data engine. The sequence annotator is one component of that engine, not the whole system.

4 Sequence detection and literature boundary

A classical YOLO annotation is a set: each object carries a class and a bounding box. Standard detectors emit this with specialized heads. ASIDE represents the same information as a token sequence.

Let an image annotation set be

$$A = \{(c_i, b_i)\}_{i=1}^{N}, \qquad b_i = (x_1, y_1, x_2, y_2), \tag{1}$$

where N varies across images, c_i is a canonical ontology class, and boxes use corner form (aligned with sequence tokens in v0.1). The learning problem narrows from Image→Annotation to

$$\text{Image} \;\rightarrow\; \text{Annotation Token Sequence}. \tag{2}$$

Pix2Seq already quantizes coordinates, jointly models class and geometry tokens, and terminates variable-length outputs with EOS [2]. Novelty for ASIDE therefore cannot start from "we cast detection as a sequence." The scientifically interesting part is the surrounding learning data engine and how memory binds into the annotator.

5 System overview

The target architecture runs from a user request to YOLO feedback. A Task Agent maps the problem into an ontology. One branch discovers public datasets; another ingests customer images. Both meet in a Schema Normalizer. A vision encoder produces embeddings; a Seq2Seq annotator proposes labels; a VectorDB retrieves similar history; grounding, SAM 2, and an existing YOLO (if any) provide second opinions. Consensus separates certain from uncertain cases. Clean data trains YOLO; an Error Agent expands the dataset. Thick boxes in the architecture are decision points; shaded paths are memory or uncertainty routes.

6 Task Agent and the agentic / deterministic split

Users should not supply technical dataset parameters. A sentence such as "I want to detect holes, cracks, and abnormal veins in marble" becomes a structured problem: domain, material, targets, tasks, optional attributes. An LLM may be used here. The design rule is explicit: we will not make everything an agent.

Agentic reasoning	query generation, ontology proposals, prioritization
Deterministic pipeline	format conversion, geometry, hashing, schema compile

Agents reason brittlely; conversion, normalization, and hashing must be reproducible code.

7 Dataset discovery, registry, and quality

From Task Agent outputs (e.g., crack, hole, cavity, marble, industrial surface defect), a Dataset Discovery Agent searches catalogs such as Kaggle and Hugging Face Hub. It does not search only "marble"; it expands queries. Marble data may be scarce; granite / stone / surface-defect data may still help transfer learning. Datasets are registered before training.

Registry fields include id, name, source, source_url, license, domain, image_count, annotation_format, resolution statistics, classes, quality_score, download_date, version, and optionally a content hash. A License Agent separately checks commercial vs. research use.

A large dataset is not automatically better than a small one. A Quality Agent produces a composite score

$$Q_d = w_D D + w_A A + w_C C + w_I I + w_L L + w_S S, \qquad \sum_k w_k = 1, \tag{3}$$

where D is domain similarity, A annotation quality, C class coverage, I image quality, L license suitability, and S sample diversity. Weights are heuristic in v0.2 and may later be learned.

8 Ontology Agent

Dataset A may say "hole" and "crack"; Dataset B "cavity" and "fracture"; Dataset C "void" and "fissure." Blind merging creates spurious classes. An ontology maps source names to canonical categories, each with a confidence. Low-confidence mappings go to human review—a "hole" may be a surface void in one corpus and a drilled hole in another.

9 Annotation schema normalization

Public data arrive as YOLO, COCO, Pascal VOC, CSV, custom JSON, polygons, or mask PNGs. All convert first to `UnifiedAnnotation`: `image_id`, `width`, `height`, and an object list with `object_id`, `canonical_class`, `source_class`, `bbox`, `polygon`, `mask`, `attributes`,

Table 1: Example ontology mappings (illustrative numbers).

Source class	Canonical	Confidence	Decision
cavity	CAVITY	0.97	auto-accept (example)
hole	CAVITY	0.71	human review (example)
fissure	CRACK	0.93	auto-accept (example)
fracture	CRACK	0.88	auto-accept (example)

`source_dataset`, `verified`, `confidence`. Kaggle, Hugging Face, customer, pseudo-label, and human-corrected data then share one schema and can be exported via YOLO or COCO exporters.

10 Vision encoder

An image $I \in \mathbb{R}^{H \times W \times 3}$ passes through a vision encoder E_θ (e.g., ViT) that yields visual tokens

$$V = E_\theta(I) \in \mathbb{R}^{N \times d}, \tag{4}$$

where N is the number of patches/tokens and d the latent width. The decoder cross-attends to V at every step.

11 Annotation tokenizer and sequence format

Floating-point boxes cannot be vocabulary items. Coordinates are normalized to $[0, 1]$ then quantized into B bins (e.g., $B = 1000$ maps $x' = 0.314$ to `<X_314>`), following Pix2Seq [2].

The v0.1 format is deliberately simple:

$$S = \big[\texttt{BOS},\ O_1,\ O_2,\ \ldots,\ O_n,\ \texttt{EOS}\big], \qquad O_i = \big[\texttt{OBJ},\ c_i,\ y_{1i},\ x_{1i},\ y_{2i},\ x_{2i}\big], \tag{5}$$

using Pix2Seq corner order.

12 EOS, variable length, and object ordering

One image may contain 1 object; another 38. There is no fixed detector-slot count: decoding ends at EOS. A second issue is that true annotations are *sets*: permutations of the same objects are equivalent under AP/IoU, yet Transformers require an order. Pix2Seq finds randomizing object order during training helpful [2]; ASIDE will ablate ordering policies.

13 Sequence model and loss

Visual tokens condition an autoregressive decoder. Factorization is

$$P(S \mid I) = \prod_{t=1}^{T} P\big(s_t \mid I,\ s_{1:t-1}\big). \tag{6}$$

Version 0.1 uses token-level cross-entropy against gold tokens s_t^*:

$$\mathcal{L}_{\text{seq}} = -\sum_{t=1}^{T} \log P\big(s_t^* \mid I,\ s_{1:t-1}^*\big). \tag{7}$$

Later versions may add box regression, IoU, and retrieval-consistency terms:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{token}} + \lambda_2 \mathcal{L}_{\text{bbox}} + \lambda_3 \mathcal{L}_{\text{IoU}} + \lambda_4 \mathcal{L}_{\text{retrieval}}, \tag{8}$$

with $\lambda_2 = \lambda_3 = \lambda_4 = 0$ in v0.1.

14 Pure Seq2Seq versus hybrid geometry

Pure Seq2Seq matters scientifically. Production may also need a hybrid: a semantic decoder (class, attributes, relations) plus a continuous geometry head (x_1, y_1, x_2, y_2). Crack severity, relations, and orientation fit sequences naturally; box regression can be more efficient with a dedicated head. The choice is an ablation, not speculation.

15 Linked sequences and temporal memory

The independent-frame model $P(A \mid I)$ is incomplete for conveyors and video. For frames $I_1, \ldots, I_t$ with annotations $A_1, \ldots, A_t$, we condition on history:

$$P(A_t \mid I_t, I_{t-1}, A_{t-1}, \ldots). \tag{9}$$

Short-term memory M_t evolves as

$$M_t = f(M_{t-1}, V_t, A_t), \qquad A_t = D(V_t, M_{t-1}), \tag{10}$$

so the model does not restart from zero each frame. SAM 2's streaming session memory is a strong existence proof for temporal memory in segmentation [5].

16 Production sequence memory

Linked sequences are not only video. The manufacturing path Raw→Cut→Polished→Drilled→Final QC is a physical-object sequence. A crack visible on raw marble may fade after polishing yet should remain remembered. This is *production sequence memory*.

17 VectorDB and Visual RAG

Short-term memory covers near history of the same product. Long-term memory answers "have I seen something like this in the last three months?" A vision encoder emits embedding $z = E(I)$ (or ROI crop); a VectorDB stores it. At inference, Top-K neighbors R_t are retrieved. VectorDB is not a classifier: neighbors say what past examples looked like; the Transformer decides.

$$R_t = \mathrm{TopK}(z_t), \qquad A_t = D(V_t, R_t). \tag{11}$$

Analogy to NLP RAG: question→embedding→VectorDB→documents→LLM→answer becomes image→vision encoder→embedding→VectorDB→similar labeled examples→annotation Transformer→annotati

18 Four memory types and retrieval attention

Long-term memory may hold four collections: **image memory** (similar scenes), **object memory** (similar ROIs), **error memory** (past mistakes), and **hard-example memory** (chronically uncertain / active-learning cases). If the system once mistook a shadow for a hole, retrieval can surface "predicted hole, human corrected shadow" context.

Rather than stacking raw neighbor images, memory cross-attention filters retrieved records:

$$z_t = \mathrm{Pool}(V_t), \quad R_t = \mathrm{TopK}(z_t; K), \quad \mathrm{Attn} = \mathrm{softmax}\Big(\frac{Q_{\text{current}} K_R^\top}{\sqrt{d}}\Big) V_R, \tag{12}$$

with suggested $K = 8$ in v0.3.

19 Core formula: short and long memory

Two memories meet at the decoder, with ontology as an additional condition:

$$A_t = D\big(V_t,\ M_t^{\text{short}},\ M_t^{\text{long}},\ O\big). \tag{13}$$

A tighter writing packs decoder context into a single state—the mathematical core of the proposal:

$$H_t = \big[\, V_t,\ T_t,\ R_t,\ E_t,\ O \,\big], \qquad A_t = \text{Decoder}(H_t), \tag{14}$$

where T_t is temporal/production memory, R_t retrieved semantic memory, E_t error / learning-history memory, and O the ontology. The measurable mechanism claim attaches here: how memory binds into the annotation model.

20 Storage architecture

Not everything lives in a VectorDB. Three stores separate concerns:

- **SQL:** datasets, metadata, ontology, annotations, model versions, corrections, audit;
- **Object storage:** images, video, masks, checkpoints;
- **VectorDB:** embedding collections for similarity search.

This is an information-architecture decision independent of any application framework.

21 Teacher models, SAM 2, and consensus

A sequence model must not unilaterally bless its own pseudo-labels. Second opinions come from a grounding detector [4], SAM 2 [5], an existing YOLO if present, and temporal memory. A produced crack box can prompt SAM 2; mask–box consistency is checked.

Acceptance is not a single softmax. The organizing principle is

$$\text{Confidence} \neq \text{Correctness}. \tag{15}$$

A multi-signal consensus score (weights calibrated; later possibly gated) is

$$C_i = w_s C_{\text{seq}} + w_g C_{\text{ground}} + w_r C_{\text{retrieval}} + w_t C_{\text{temporal}} + w_m C_{\text{seg}} + w_a C_{\text{agreement}}, \tag{16}$$

$$\text{Reliability} = f(\text{agreement},\ \text{grounding},\ \text{retrieval},\ \text{temporal},\ \text{history}). \tag{17}$$

22 Humans, active learning, and error memory

Claiming "100% automatic labeling" is especially risky for industrial data. The correct objective is reducing Human Labeling Cost. Conceptually, of 100,000 images one might auto-accept ∼82,000, audit ∼13,000, and manually review ∼5,000—humans only touch the hard queue. Ratios are targets, not measurements.

Review priority is not random:

$$P_i = \alpha U_i + \beta D_i + \gamma H_i + \delta N_i, \tag{18}$$

with uncertainty U_i, teacher disagreement D_i, historical difficulty H_i, and novelty N_i (low retrieval similarity). Per-example learning history

$$H_i = (\text{confidence hist.},\ \text{prediction hist.},\ \text{correction hist.}) \tag{19}$$

feeds difficulty and error memory. A single human correction (HOLE→SHADOW) can update SQL, verified embeddings, error memory, and the training buffer at once.

23 Hard negatives and the self-improving loop

If the model confuses shadows with holes, naïve training asks for more holes. ASIDE asks for hole-*like* non-holes: shadows, dark veins, oil marks, reflections, dark discoloration—a hard-negative set that typically cuts industrial false positives.

The loop is

$$D_t \;\rightarrow\; M_t \;\rightarrow\; E_t = \text{Evaluate}(M_t) \;\rightarrow\; D_{t+1} = D_t + \Delta D(E_t). \tag{20}$$

The dataset is not a static file; it evolves from error signals. Hence *self-improving data engine*.

24 Domain shift and customer data

Kaggle images may have perfect lighting and fixed cameras; factories have sunlight, dust, water, motion blur, and different materials. Public vs. factory embedding distributions are compared (MMD, Fréchet, or centroid cosine—chosen in v0.3). Large distance means public data is bootstrap-only. Over time,

$$D^{\text{verified}}_{\text{customer}} \tag{21}$$

becomes the valuable asset; public catalogs are a ladder, not a permanent dependency.

25 Rich outputs, relations, and the role of YOLO

Sequence outputs need not stop at boxes. Future tokens may include severity, length, orientation, relations, and status. YOLO remains strong on class+geometry; the sequence layer can carry attributes, relationships, context, and history. Example relations after detection: `drill_point_7 INTERSECTS crack_1`, `NEAR vein_2`—inputs to planning / process control.

Critically, the Seq2Seq model need not be the production detector. A heavy offline data engine can produce excellent datasets; a light YOLO (e.g., TensorRT) runs at line rate. Factory logs—predictions, false positives, operator corrections, rare objects—flow back:

$$\text{Factory YOLO} \rightarrow \text{errors} \rightarrow \text{data engine} \rightarrow \text{new annotations} \rightarrow \text{retrain} \rightarrow \text{new YOLO}.$$

26 Relation to prior work

Pix2Seq places detection in a language-modeling frame [2]. Florence-2 unifies detection, grounding, segmentation, and captioning under prompt+seq2seq and describes automated annotation with iterative refinement for FLD-5B [7]. Grounding DINO supplies open-set / text-guided detection [4]. SAM 2 supplies promptable segmentation and temporal streaming memory [5]. We therefore cannot claim invention of these components in isolation. ASIDE differentiation must lie in agentic retrieval/memory, industrial specialization, and annotation feedback.

Main scientific risk. The system may look clever yet reduce to scripts around Grounding DINO + SAM + VectorDB + LLM + YOLO. That can be a useful product without being research. Research requires a new, measurable mechanism for how memory binds into the annotation model—especially around Eq. (14).

27 Research question, originality, and ablation

The research question is no longer "let us emit YOLO labels with Seq2Seq." Stronger form:

> Can retrieval-augmented visual memory, interconnected sequences, model disagreement, and learning-history signals reduce human labeling load and improve annotation quality for domain-specific detection?

Table 2: Literature components and attachment points in ASIDE.

Component	**Source**	**Role in ASIDE**
Detection as sequence	Pix2Seq [2]	v0.1 annotator skeleton
Unified seq2seq + data engine	Florence-2 [7]	task unity / iterative annotation inspiration
Open-set / text-guided det.	Grounding DINO [4]	teacher / second opinion
Promptable seg. + video memory	SAM 2 [5]	mask validation and temporal example
Line-rate detector	YOLO family [6]	online production model

The combination under study is sequence annotation with retrieval, temporal memory, error memory, consensus, active learning, and detector feedback. Individual pieces exist in the literature; measuring the combination and the memory binding is our job.

Ablations isolate contribution:

Table 3: Ablation matrix. Version 0.1 targets only experiment A. Metrics: annotation AP, human correction %, FP, FN.

Exp.	Seq	VectorDB	Temporal	Error mem.	Teacher	Question
A	+					Is the sequence annotator viable?
B	+	+				Does retrieval raise annotation AP?
C	+	+	+			Does temporal binding cut FN?
D	+	+	+	+		Does error memory stop repeats?
E	+	+	+	+	+	Does consensus cut human correction %?

28 Version plan and the v0.1 experiment

The full system is large. Shipping everything at once hides what works. Versions are deliberately narrowed:

v0.1	Image→annotation sequence model (this document's scope)
v0.2	Ontology + schema normalization + quality scoring
v0.3	VectorDB / Visual RAG + domain-shift metrics
v0.4	Teacher consensus + reliability gating
v0.5+	Temporal / production memory, active learning, YOLO loop
v1.0	End-to-end Discover→Improve operating cycle

Next technical work should start from tensor shapes: encoder → tokenization → decoder → loss → training loop → inference. Productization (UI, auth, queues) is out of scope until $H_t \to$ Decoder is measurable.

29 Evaluation and success criteria

Downstream YOLO mAP alone is insufficient. Two systems are measured separately.

Annotation engine. Annotation precision/recall, box IoU, mask IoU, class accuracy, human correction rate, auto-accept error rate, human minutes per 1,000 images.

Downstream detector. mAP50, mAP50:95, precision, recall, per-class AP, false-positive rate, false-negative rate.

Business metric. Human Annotation Time (hours per 1,000 or 10,000 images), tracked independently of mAP. Target scenario (not a result): 10,000 images from 80 h to 8 h review would be a 90% workload reduction—often more meaningful than "mAP +0.7" alone. This number is *unmeasured* in v0.1.

30 Conclusion

At this level of definition, ASIDE is no longer "a Seq2Seq project that eases YOLO labeling." It is an autonomous CV data-production system that combines a Seq2Seq annotator, short-term temporal memory, VectorDB long-term visual memory, active learning, and downstream detector feedback. In narrative form for v1.0 (not the v0.1 scope): the customer states a natural-language factory goal; the system builds an ontology, finds public data, scores quality, normalizes schemas, trains a sequence annotator, encodes unlabeled images, binds similar history and temporal memory, proposes labels, verifies with teachers, auto-accepts high reliability, routes uncertainty to humans, writes corrections into error memory, trains YOLO, observes production errors, and continues the loop.

Immediate next step: make the $H_t \to$ Decoder binding in Eq. (14) experimentally measurable.

A Symbol glossary

Symbol	Meaning
I, I_t	Input image / frame at time t
A, A_t	Annotation (set or sequence)
S, s_t	Token sequence and t-th token
V, V_t	Visual tokens
z, z_t	Retrieval embedding
E_θ, E	Vision / embedding encoder
D	Annotation decoder (also dataset in the loop notation)
O_i, b_i, c_i	Object, box, class
B	Number of coordinate bins
Q_d	Dataset quality score
C_i, Reliability	Consensus / accept score
P_i	Active-learning priority
M^{short}, T_t	Short-term / temporal memory
M^{long}, R_t	Long-term retrieved memory
E_t, H_i	Error / learning-history
H_t	Unified decoder state
O	Ontology
D_t, M_t, E_t	Dataset, model, error (loop)

B Agents and services

Some units below are true LLM agents; others are ordinary services. The split follows Section 6.

Unit	Type	Responsibility
Task Agent	agent	Formalize natural-language problems
Dataset Discovery	agent	Catalog search + query expansion
Dataset Quality	service/agent	Produce Q_d; drop below threshold
License Agent	service	Commercial / research license check
Ontology Agent	agent+human	Class mapping; low confidence to human
Schema Compiler	deterministic	YOLO/COCO/VOC → UnifiedAnnotation
Embedding Service	service	Image / object embeddings
Retrieval	service	Top-K from VectorDB
Sequence Annotator	model	Emit annotation sequences
Grounding Service	model	Open-set second opinion
Segmentation Service	model	SAM 2 mask validation
Temporal Memory	service	Same-product / frame history
Consensus Engine	det.+calib.	Fuse signals into Reliability
Active Learning	service	Order human queue by P_i
Hard Negative Agent	service	Collect misleading negatives
Training Service	service	Train annotator / YOLO
Evaluation Agent	service	AP, recall, error types
Feedback Agent	agent/service	Turn errors into new data tasks

C Selected reference anchors

This list is not a full literature survey. It anchors ASIDE's position. Version 0.1 claims not a new detector invention, but a plan to measure how memory binds into the annotation model [1–7].

References

[1] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*, 2020.

[2] Ting Chen, Saurabh Saxena, Lala Li, David J. Fleet, and Geoffrey Hinton. Pix2Seq: A language modeling framework for object detection. In *International Conference on Learning Representations (ICLR)*, 2022.

[3] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[4] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision (ECCV)*, 2024.

[5] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Jiabo Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.

[6] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[7] Bin Xiao, Haiping Wu, Weijian Xu, Xiyang Dai, Houdong Hu, Yumao Lu, Michael Zeng, Ce Liu, and Lu Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.